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Grover-Inspired Evidence Selection for Document Retrieval: Relevance Constraints, Failure Analysis, and Resource Accounting

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ABSTRACT When a retrieval system returns a small set of documents, relevance and redundancy can pull selection in different directions. We study that tradeoff with a Grover-inspired pipeline: classical retrieval constructs the candidates, a cross-encoder scores relevance, and ideal amplitude amplification is simulated on a central processing unit. The base selector chooses eight of fifteen candidates. Refinement retains the first five results and searches an expanded pool for a less redundant three-document tail. Our evidence comprises completed evaluations on 1,271 test queries from scientific, nutrition-related, and financial collections; no new benchmark rankings were collected. Normalized discounted cumulative gain at rank five is 0.6878 on SciFact, 0.3689 on NFCorpus, and 0.3575 on FiQA. For the first two collections, refinement lowers mean redundancy by 3.21 and 2.63 percent, while relative recall changes by -0.93 and +1.95 percent. FiQA uses the earlier base configuration; its separately tested refinement failed. Candidate coverage, query-level losses, seed stability, and request costs add detail to the aggregate results. Matched greedy and exact refiners achieve the same reported aggregate quality, while neural inference dominates measured online time. A separate completed resource study examines reversible constraints and coefficient precision within explicit simulation limits. The results characterize a constrained relevance–redundancy tradeoff and its resource costs, but do not establish physical quantum acceleration or improved generated answers.

INDEX TERMS Amplitude amplification, cross-encoders, document retrieval, evidence selection, quantum algorithm simulation, reproducibility.

I. INTRODUCTION

AN eight-passage limit turns retrieval into a question about the whole list. A ninth passage might add useful evidence, or it might repeat material already selected. Scoring each passage for relevance does not settle that choice. The issue arises in retrieval-augmented generation because the selected material must fit the generator's context window [1]; it also arises in document search when several plausible results cover the same ground.

Selecting dissimilar passages is not a substitute for finding the right evidence. A passage may resemble another and still contain the only correct support for a claim; a dissimilar passage may be irrelevant. We distinguish relevance, redundancy, coverage, and answer quality throughout the paper. Our experiments measure judged retrieval effectiveness and an embedding-based proxy for redundancy. The proxy is not a human evidence judgment, and it cannot establish the quality of answers we did not generate.

Our selection step is motivated by quantum search, but its theoretical and measured costs have different meanings. Grover search and amplitude amplification improve oracle-query complexity under their stated access assumptions [2], [3]. A document pipeline still has to construct representations, retrieve candidates, score them, and encode an objective. It also incurs the cost of simulating amplitudes classically. That measured latency is not the complexity of an ideal black-box quantum oracle.

This article extends the accepted Qubit Chat conference manuscript [4]. The earlier work reported retrieval comparisons, an L6-guided Grover Adaptive Search (GAS) selector, protected-prefix refinement, mathematical bounds, and a bounded native-oracle resource study. Those results are prior

work. Using the completed evidence, we ask where relevant documents leave the candidate pipeline, which queries gain or lose despite the mean, how outcomes vary across five stochastic seeds, and which stages account for the full request time.

The base policy selects eight documents from a hybrid shortlist of fifteen. It uses a cross-encoder for relevance and clipped pairwise embedding cosine for redundancy. Refinement holds the first five baseline results fixed, then searches an expanded pool within the original dense top 64 for a three-document tail. The tail must satisfy score constraints, and an update must strictly reduce unquantized redundancy. This design preserves normalized discounted cumulative gain at rank five (nDCG@5), while leaving recall and lower-ranked relevance free to change.

We report completed tests on SciFact, NFCorpus, and FiQA. The SciFact and NFCorpus policies include protected refinement; the FiQA row uses the earlier base policy. Their distinction matters because the later FiQA refinement failed, and NFCorpus continued after a failed confirmation. Earlier test outcomes informed development, so the follow-up tests are exploratory rather than an untouched confirmatory assessment.

The journal extension adds four analyses of the saved results. Candidate-pool coverage and the attainable Recall@8 ceiling show what is lost before selection. Query-level gains, losses, and seed variation reveal how outcomes differ beneath the means. Reconstructed request latency separates relevance scoring from base selection and refinement. We connect those diagnostics to the recorded precision and resource grid and map the values to sealed artifacts. No new benchmark rankings or datasets were collected, and no new neural inference was performed.

The results support constrained set selection as a way to reduce measured redundancy without changing a protected ranking prefix. Recall, however, depends on the collection and on errors in the relevance model. Matched classical optimizers reach the same reported aggregate quality, placing that result with the shared formulation rather than with Grover search alone. The resource study describes an implementation and its limits; it does not establish an operational quantum advantage.

The next sections place the study in retrieval and quantum optimization, set out the formulation and protocol, and report the recorded effectiveness results. We then present the journal diagnostics and distinguish compact simulation from native reversible-oracle synthesis. The discussion addresses validity and the experiments needed for stronger deployment claims.

II. RELATED WORK

A. RELEVANCE ESTIMATION AND EVIDENCE-SET SELECTION

The pool presented to a selector depends on the retrieval signal. BM25 matches terms [5], whereas sentence encoders provide reusable query and document vectors [6]. Reciprocal-rank fusion (RRF) combines rankings without requiring their scores to share a calibrated scale [7]. We use these approaches to construct candidates and as classical comparisons; their costs are included in complete request time.

Cross-encoders process the query and passage jointly, rather than comparing independent vector encodings [8], [9]. Our embedding encoder and cross-encoder both belong to the MiniLM family [10], but the former supplies normalized vectors and the latter produces query–passage logits after MS MARCO passage-ranking training [11]–[13]. Those logits are neither calibrated relevance probabilities nor judgment labels. Higher logits therefore need not improve judged recall.

The broader ranking literature includes dense passage retrieval [14], late interaction [15], and sequence-to-sequence scoring [16]. Generative ranking agents extend the alternatives [17], while comparative reranker studies examine architecture, training overlap, generalization, and cost [18]. These are candidates for future controls, not results measured in our experiment. Our neural comparisons are limited to the pinned L2 and L6 cross-encoders on the declared pools.

RAG evaluation can separate indexing, retrieval, reranking, similarity, generation quality, and system cost [19]. We make a similar distinction: our saved experiments concern retrieved evidence and selection resources, not generated-answer quality, which would require its own evaluation.

Set-selection methods also differ in what they compare. Maximal marginal relevance (MMR) selects documents sequentially, balancing relevance against similarity to earlier choices [20]; our objective uses average pair similarity over a complete fixed-size set. Determinantal point processes provide a probabilistic model of diverse subsets [21]. They are context here, not an evaluated baseline. Table 1 separates these interaction models from the mechanisms that search them.

B. QUANTUM SEARCH AND REVERSIBLE RESOURCE ACCOUNTING

Amplitude amplification increases the probability of marked states; it does not rank equally marked documents by relevance [2], [3]. Randomized schedules address an unknown marked-state count [23], and minimum finding updates an incumbent threshold [24]. GAS applies these ideas to constrained polynomial binary optimization [22]. Our subset objective uses that framework to represent interactions among selected documents.

Work on GAS considers both schedule efficiency and constants hidden by asymptotic query complexity [25], as well as initialization, rotations, and gate construction in particular applications [26]. An oracle call has a cost determined by its encoded task. Our measured oracle performs cardinality and tail-score checks, computes signed improvement, and uncomputes; the compact simulator uses another representation and cost boundary.

Structured fixed-weight preparation is possible, as deterministic Dicke-state constructions demonstrate [27]. It is not, however, the measured cost of software using generic preparation. Matrix-product-state simulation depends on en-

TABLE 1. Methodological positioning. "Measured" identifies local controls; literature-only rows do not assert an empirical comparison.

Approach	Relevance signal	Treatment of document interactions	Relationship to this evaluation
BM25 / dense / RRF [5]–[7]	Lexical, vector, or fused ranks	Candidate scoring; no joint redundancy objective	Measured classical policies and upstream features
L2/L6 cross-encoders [9], [13]	Joint query–passage relevance	Scores documents independently	Measured on declared dense or hybrid pools
MMR [20]	Estimated relevance	Sequential novelty penalty	Measured cosine/L6 variants with frozen tuning
Determinantal selection [21]	Quality-weighted similarity model	Probabilistic subset diversity	Literature context; not an evaluated baseline
Generative reranking [17], [18]	Language-model ranking judgments	Model-dependent listwise interactions	Literature context; no local measurements
Classical matched optimizers	Identical saved L6 features	Same set objective and admissibility rules	Greedy, exact float/integer, and random controls
GAS [22]	Encoded polynomial objective	Adaptive improvement search	Adopted search framework; no new GAS algorithm
Protected-prefix policy [4]	Original and additional L6 logits	Fixed head, constrained tail, pair redundancy	Prior formulation; expanded analysis in this article

tanglement [28]; success on a small circuit does not establish feasibility for a larger one. Access models and implementation assumptions also matter when interpreting quantum algorithms [29]. We keep those boundaries in the resource analysis.

C. EVALUATION PRACTICE AND EXTENSION OVER PRIOR WORK

BEIR provides heterogeneous collections and standardized retrieval tasks [30]. Our selected collections address scientific claims, nutrition-related search, and financial questions through SciFact, NFCorpus, and FiQA, respectively [31]–[33]. Their judgment distributions differ. Candidate coverage and the number of relevant documents per query therefore matter when interpreting absolute recall; we examine coverage within each collection.

We use gain-based ranking measures and independent implementations to support consistent evaluation [34], [35]. For paired inference, queries are the observations, and the family of method comparisons must be considered [36]–[38]. The recorded statistical families and earlier hypotheses remain intact; the added diagnostics are descriptive. Relative to the conference manuscript [4], the new empirical material analyzes saved outcomes rather than adding test collections or a newly tuned policy.

D. APPLICATION-SPECIFIC DECISION PIPELINES

Related decision pipelines have been studied in other applications. Blockchain-mining work covers trust scoring [39], [40], VIKOR-based ranking [41], reputation and consensus [42], [43], and miner selection or classification [44], [45]. Other studies address multimodal waste detection [46], smart-contract repair [47], financial-fraud detection [48], plant-disease recognition [49], traffic forecasting [50], and obscured-fruit classification [51]. We cite them as application context, not as results from our retrieval evaluation.

III. PROBLEM FORMULATION AND SYSTEM BOUNDARIES

A. FROM DOCUMENTS TO A LIMITED EVIDENCE BUDGET

Let q denote a query and D a collection of parent documents. Preprocessing splits each parent into one or more chunks. The best-matching chunk supplies both the parent's dense-retrieval score and its representative passage for relevance inference. Judgments apply to the parent, however; a relevant document need not have relevant content in every chunk.

Write $\mathcal{C}_{64}(q)$ for the dense top-64 parents, $\mathcal{C}_{15}(q)$ for the frozen hybrid shortlist, and S for a selected set of $k = 8$ distinct parents. Evaluation orders the selected parents using a ranking $\pi(S)$. The evidence budget limits $|S|$, not the token count of a generated answer; the reported retrieval benchmark invokes no answer generator.

For candidate i, the classical pipeline produces query cosine c_i, lexical score b_i, normalized representative vector v_i, and cross-encoder logit z_i. Let

$$d_{ij} = \min\{1, \max\{0, v_i^\top v_j\}\}, \qquad i \neq j. \tag{1}$$

We measure set redundancy as

$$\mathcal{R}(S) = \frac{1}{\binom{k}{2}} \sum_{\{i,j\} \subset S} d_{ij}. \tag{2}$$

The measure ranges from zero to one. It is the mean clipped cosine, not a percentage of duplicate documents. Negative cosine is clipped to zero, so a zero penalty does not establish that passages are semantically independent.

The base selector maximizes a relevance–redundancy utility. Refinement poses a different problem: minimize redundancy while satisfying protected-prefix and neural-score conditions. Neither directly optimizes the unknown test judgments. We use judgments after saving rankings, including in the retrospective journal diagnostics.

B. CANDIDATE OPPORTUNITY AND A RECALL CEILING

For a query with a nonempty set J_q of positively judged relevant parents and a candidate pool C, we define judged coverage and a cardinality-aware recall ceiling by

$$\mathrm{Cov}(C,q) = \frac{|C \cap J_q|}{|J_q|}, \tag{3}$$

$$\mathrm{Ceil}_8(C,q) = \frac{\min\{8, |C \cap J_q|\}}{|J_q|}. \tag{4}$$

An eight-document output drawn from C cannot have Recall@8 above $\mathrm{Ceil}_8(C,q)$. It cannot contain more than eight relevant documents or more relevant documents than the pool contains. The diagnostic equality is attainable by selecting as many judged-relevant pool members as the budget permits, but a practical system is not assumed to identify them without judgments.

Both quantities are nondecreasing when one pool contains another. Expansion can create more opportunity for judged recall without improving a particular selector's output. An optimizer confined to a pool cannot recover a relevant parent absent from it. This separates losses between corpus and pool from losses between pool and selected set, and from ordering within that set.

The ceilings use observed judgments; a useful document may be unjudged. They bound the official judged metric, not every possible notion of relevance. The expanded-pool ceiling also relaxes the refinement's prefix and score constraints, making it optimistic for the constrained task. A gap to that ceiling cannot be assigned to the search algorithm alone.

C. IMPLEMENTATION AND MEASUREMENT BOUNDARIES

Figure 1 separates retrieval, neural relevance inference, base selection, and optional refinement. The application also provides document-ingestion and question-answering interfaces, but the experiments do not measure interface quality, optical character recognition accuracy, or live generated answers. The research pipeline uses public parent-document judgments and frozen passage construction.

The main selector supplies the retrieval rankings using amplitudes for subsets of the required size and classically constructed objective values. The native study constructs reversible circuits with explicit decision and workspace registers for bounded synthesis and validation. Neither model includes device routing, noisy execution, or measurements on a quantum processor.

IV. RELEVANCE-GUIDED SUBSET SEARCH

A. HYBRID CANDIDATE CONSTRUCTION

The hybrid shortlist is drawn from the dense top-64 pool. Within that pool, let $c_{\max} = \max_i c_i$ and $b_{\max} = \max_i b_i$. The reproduced score is

$$h_i = (1-\beta)\frac{1+c_i}{1+c_{\max}} + \beta\frac{b_i}{b_{\max}}. \tag{5}$$

When lexical scores are negligible, only the dense component is used; when its denominator is negligible, that component becomes $(1+c_i)/2$. The numerical tolerance is 10^{-12}. Hybrid score determines the top-fifteen shortlist, with document ID breaking ties. Thus fusion operates on an already retrieved pool, rather than searching the collection quantum mechanically.

Hybrid scores and dense cosine do not replace the raw cross-encoder logits. For the original fifteen candidates, normalized relevance is

$$r_i = \frac{z_i - z_{\min}}{z_{\max} - z_{\min}}, \tag{6}$$

with $r_i = 0.5$ when the range is negligible. Negative logits are permitted. The normalized values supply optimization coefficients; the final relevance order uses raw logits. Since normalization is local to a shortlist, its values are not calibrated probabilities that can be compared across pools.

B. BASE OBJECTIVE AND INTEGER ENCODING

For $|S| = k = 8$, we define base utility as

$$U(S) = \frac{1}{k}\sum_{i\in S} r_i - \lambda \mathcal{R}(S). \tag{7}$$

Both terms depend on classical features. Using $A_b = 2^8 - 1 = 255$, the rounded coefficients are

$$a_i = \mathrm{round}(A_b r_i), \tag{8}$$

$$a_{ij} = \mathrm{round}\left(\frac{2A_b\lambda}{k-1} d_{ij}\right), \tag{9}$$

$$E_b(S) = -\sum_{i\in S} a_i + \sum_{\{i,j\}\subset S} a_{ij}. \tag{10}$$

Lower integer energy is preferred, and rounding uses ties to even. The raw-L6 top eight initialize the incumbent. Selected outputs are ordered by descending raw logit, with document ID resolving ties.

Without rounding, $-E_b(S)/(A_b k)$ equals (7). Each of the k unary coefficients and $\binom{k}{2}$ pair coefficients has rounding error of at most one half. It follows that

$$\left|\frac{-E_b(S)}{A_b k} - U(S)\right| \le \frac{k+1}{4A_b}. \tag{11}$$

This bound applies to coefficient approximation for a fixed set. It says nothing about relevance-model error or the gap between utility and judged effectiveness. At $\lambda = 0$, the raw-relevance incumbent already maximizes relevance-only utility, so the implementation retains it rather than changing membership when quantized energies tie.

C. PROTECTED-PREFIX REFINEMENT

Let the seed-specific base ranking be $(s_1, \ldots, s_8)$, its ordered head be $H = (s_1, \ldots, s_5)$, and its original tail be $T_0 = \{s_6, s_7, s_8\}$. Construct an expanded candidate union

$$\mathcal{C}_\cup = \bigcup_{\beta\in\{0,0.25,0.5\}} \mathrm{Top}_{15}(h_\beta; \mathcal{C}_{64}). \tag{12}$$

The union contains the original shortlist and at most 45 parents. We retain the original-shortlist logits and score only

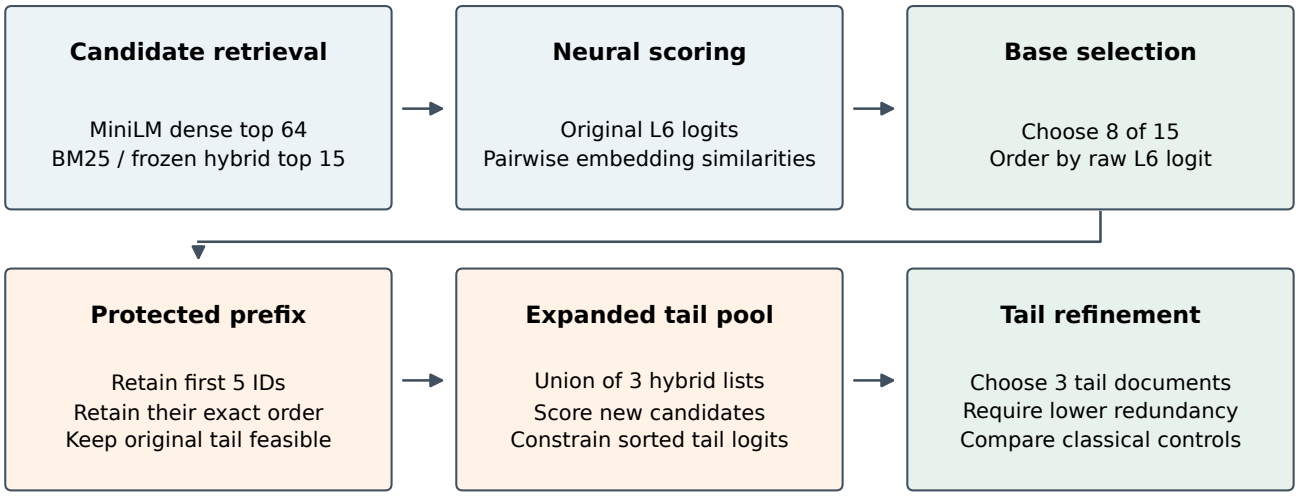


FIGURE 1. System boundaries and reported configurations. Classical retrieval and neural inference precede ideal amplitude amplification simulated on a CPU. The separate native resource study does not replace those classical stages or constitute a full hardware execution of the retrieval pipeline.

newly introduced candidates. This avoids changing the baseline through a different inference-batching context.

The refinement searches $T \subseteq \mathcal{C}_{\cup} \setminus H$ with $|T| = 3$. Sort the tail logits in ascending order, denoted $z_{(1)}(T) \le z_{(2)}(T) \le z_{(3)}(T)$. Admissibility requires

$$z_{(j)}(T) \ge z_{(j)}(T_0) - \varepsilon, \quad j \in \{1,2,3\}, \quad \varepsilon = 10^{-12}. \tag{13}$$

The constraint compares sorted score values, not document identities: an original tail document may be replaced. Candidates below the smallest permissible score are removed before constructing the subset basis. If M candidates remain, its cardinality basis has $\binom{M}{3}$ states, some of which may still fail the other score conditions.

For fixed thresholds $\theta_j = z_{(j)}(T_0) - \varepsilon$, condition (13) is equivalently expressed by

$$\sum_{i \in T} \mathbf{1}\{z_i \ge \theta_j\} \ge 4 - j, \qquad j = 1,2,3. \tag{14}$$

In an ascending tail, the jth value reaches θ_j exactly when at least the largest $4-j$ values reach that threshold. The inclusive comparison handles repeated logits. This count form suits reversible implementation; the sorted form makes admissibility easier to interpret.

The refinement minimizes $\mathcal{R}(H \cup T)$. With $A_r = 2^{16} - 1$, its integer objective is

$$E_r(H \cup T) = \sum_{\{i,j\} \subset H \cup T} \mathrm{round}(A_r d_{ij}). \tag{15}$$

Reported redundancy includes head–head, head–tail, and tail–tail terms, even though the head–head contribution is constant during refinement. The oracle marks admissible states whose integer energy is strictly below the incumbent's. The controller accepts a measured update only if true, unrounded redundancy improves by more than ε.

We return the original head in its existing order, followed by the selected tail sorted on raw L6 logits. A high-logit new candidate therefore cannot displace a protected head document. Global union-L6 serves as the control with a different ordering rule.

D. WHAT THE CONSTRAINTS GUARANTEE

Proposition 1 (Protected ranking metric): For fixed query judgments, preserving the first five document IDs and their order preserves nDCG@5 exactly, independently of how the last three outputs are chosen.

Proof: The numerator of nDCG@5 contains only the gains and discounts at those five ranks. Its ideal denominator is a function of the fixed judgments, not the retrieved tail. Both quantities are unchanged. The equality therefore also holds after seed and query averaging. ■

Proposition 2 (Incumbent safety): The original tail is admissible and survives the score prefilter. Every accepted refinement retains eight unique outputs and never increases the reported redundancy relative to its starting base set.

Proof: Each sorted original-tail score meets its own bound in (13), including ties within the tolerance. Its docu-

ments therefore survive the smallest-threshold prefilter and supply a cardinality-feasible initial state. The controller replaces that state only with a valid measured tail of size three disjoint from the protected head and with strictly lower true redundancy. Induction over accepted updates proves the claim. If the search ends without improvement, its feasible incumbent is still returned. ■

The set can remain unchanged if the remaining basis is trivial, the incumbent is optimal, or sampling does not find an improvement. The guarantees also leave recall unprotected. An irrelevant candidate can have a higher cross-encoder score and lower pair similarity than a relevant original-tail document. Replacing the latter satisfies the neural-score constraints but reduces the number of judged-relevant outputs. This failure appears in the saved results in Section VII.

For general pair precision b, the refinement scale is $A = 2^b - 1$. Rounding each of the 28 pair coefficients yields

$$\left| \frac{E_r(S)}{28A} - \mathcal{R}(S) \right| \leq \frac{1}{2A}. \tag{16}$$

For exact integer minimizer S_A and exact float minimizer $S_\star$ over the same admissible domain, applying this inequality twice gives

$$\mathcal{R}(S_A) - \mathcal{R}(S_\star) \leq \frac{1}{A}. \tag{17}$$

The budgeted GAS output need not be the exact encoded optimum. Equation (17) therefore does not bound its search error. At sixteen bits, the per-set approximation bound is about 7.63×10^{-6}; greater precision cannot correct a relevance score that misjudges a document.

E. IDEAL AMPLITUDE AMPLIFICATION AND STOPPING RULES

The base cardinality basis contains $F_b = \binom{15}{8} = 6{,}435$ states. Refinement uses $F_r = \binom{M}{3}$ states. Write $|u\rangle = F^{-1/2} \sum_x |x\rangle$ for the uniform state on the corresponding cardinality basis. An oracle and reflection step are

$$O_\tau |x\rangle = (-1)^{[x \text{ admissible} \wedge E(x) < \tau]} |x\rangle, \tag{18}$$

$$G_\tau = (2|u\rangle\langle u| - I) O_\tau. \tag{19}$$

The compact simulator executes the sign changes and reflection about the mean explicitly, then draws a state according to squared amplitudes. These ideal quantum-search operations run on a classical CPU. The controller does not choose an ideal rotation number from a precomputed optimum or marked-state count.

Each stage starts from a feasible incumbent with sampling window $m = 1$. Within the remaining phase budget, a trial draws an integer from zero to $\lceil m \rceil - 1$, prepares the uniform state, applies that many steps, and measures. A strict accepted improvement resets the window; otherwise it grows by 1.2 up to $\sqrt{F}$. For each stage and seed, the recorded limits are 64 phase calls, 32 trials, and a cooperative 15-second time limit. Base and refinement use separate budgets, so a refined request can use up to 128 phase calls.

Table 2 exposes a cost outside the phase-call allowance: the compact simulator evaluates every basis state before amplification. Exact enumeration can inspect that same prepared objective array without simulating amplitudes. A low logical phase count consequently does not establish lower wall-clock work for this selector.

V. DATA AND EXPERIMENTAL PROTOCOL

A. COLLECTIONS, SPLITS, AND MODEL IDENTITIES

Table 3 records the collections, complete judged test-query counts, and frozen base settings. Every positive-query ID in each official test split is evaluated: 300 SciFact, 323 NFCorpus, and 648 FiQA queries. We exclude the partially collected fourth collection. The separate resource study uses saved features from the same domains, not an additional retrieval-effectiveness test.

The dense encoder, `sentence-transformers/all-MiniLM-L6-v2`, produces normalized 384-dimensional vectors. For relevance we use `cross-encoder/ms-marco-MiniLM-L6-v2`, with an L2 cross-encoder as the smaller neural control. Repository revisions and model-file hashes in the evidence manifest identify the artifacts used. The journal analysis downloads no new weights and reruns no inference.

We normalize document text with Unicode NFKC and whitespace collapse, using deterministic text handling. Tokenizer offsets are computed without padding or truncation. Chunks contain at most 190 WordPiece content tokens and advance by 158 content tokens; adding two special tokens gives a model input of at most 192 tokens. The maximum query cosine over a parent's chunks becomes its score, with the lowest chunk index breaking an exact tie. We use that original chunk as the representative passage, so methods sharing a pool also share parent IDs and passage representations.

Query lengths must satisfy the content-token limit, and query–passage pairs must satisfy the cross-encoder's 512-token limit; neither check permits silent truncation. The original fifteen logits are retained during expansion so that a different batching context cannot perturb the baseline. Hashes of the corpus, queries, and judgments link the saved inputs to the recorded runs.

B. DEVELOPMENT DECISIONS AND FOLLOW-UP STATUS

For SciFact, the fixed development sample comprises one fifth of the official training queries; NFCorpus and FiQA use their supplied development splits. The base search considers

$$\begin{aligned} \beta &\in \{0, 0.25, 0.5\}, \\ \lambda &\in \{0, 0.01, 0.025, 0.05\}. \end{aligned} \tag{20}$$

All stochastic settings use seeds zero through four. Eligibility requires mean nDCG@5 and Recall@8 to match or exceed L6-only on the same shortlist within 10^{-12}. Among eligible settings, the lexicographic order first maximizes nDCG@5, then Recall@8; it then minimizes redundancy, followed by λ and finally β.

SciFact selects a zero diversity weight, which the rule permits. At that setting the relevance-ranked base incumbent al-

TABLE 2. Refinement controller in implementation order. Integer marking and true-redundancy acceptance have different roles; neither uses test judgments.

Require: Original ranked eight B, expanded scored pool C, seed, and recorded stage budgets
Set $H \leftarrow B[1{:}5]$; $T_0 \leftarrow B[6{:}8]$; incumbent $T \leftarrow T_0$; window $m \leftarrow 1$
Preserve original L6 logits; build inclusive score thresholds from T_0
Remove tail candidates below the smallest threshold; enumerate the size-three compact basis
Construct integer energies and admissibility flags classically for that basis
while phase, trial, and cooperative time limits permit another trial **do**
  Draw a rotation count from the current window, restricted by the remaining phase allowance
  Prepare $|u\rangle$; phase-mark admissible integer improvements; reflect; measure a candidate tail T'
  if T' is admissible, has lower integer energy, and $\mathcal{R}(H \cup T') < \mathcal{R}(H \cup T) - \varepsilon$ **then**
    Accept $T \leftarrow T'$ and reset $m \leftarrow 1$
  else
    Grow $m \leftarrow \min\{1.2m, \sqrt{F}\}$
  end if
end while
return Original ordered H, followed by T sorted by raw L6 score; retain counts and termination status

TABLE 3. Recorded datasets and configurations. Documents are parent documents. SciFact development is derived from official training; the other two use official development splits. The lexical weight is β and the base redundancy weight is λ.

Dataset	Documents	Dev.	Test	β	λ	Reported policy
SciFact	5,183	161	300	0.25	0.000	Base + refinement
NFCorpus	3,633	324	323	0.50	0.050	Base + refinement
FiQA	57,638	500	648	0.00	0.025	Base

ready optimizes the unquantized relevance objective. Calling the search GAS does not by itself imply a nonzero diversity effect; the objective, frozen weights, and search procedure have to be interpreted separately.

The L6 MMR grid is 0, 0.1, 0.25, 0.5, 0.75, and 1: six values. The recorded relevance-first preference selects zero diversity weight on these collections. Its measured row represents one operating point, not the complete MMR tradeoff frontier. Cosine-guided controls retain their own frozen settings. Legacy threshold selection considers $\{0.3, 0.5, 0.7\}$ using development nDCG@5 and the declared tie preferences.

The deterministic 100-query confirmation samples exclude historical development, pilot, and test IDs. The descriptive gate compares refinement with its matched base: mean nDCG@5 must be equal, the mean-recall ratio at least 0.98, and mean redundancy strictly lower. NFCorpus lost 2.1561% relative recall in confirmation and failed the gate. A recorded decision continued to the full test without changing the method or global gate; the later outcome does not turn the confirmation into a pass.

Earlier test outcomes had been observed during development. Although the reported parameters were not retuned against the newer follow-up judgments, those official tests are not an untouched holdout for the redesign. The journal diagnostics retrospectively describe the same saved runs.

C. CONTROLS AND COMPARISON FAIRNESS

The original evaluation has twenty methods, with seven expanded-pool policies added for protected-refinement runs. The controls include dense cosine, dense/BM25 RRF with rank constant 60, native cosine and L6 MMR, cosine-guided GAS, L6-only ranking, base L6-guided GAS, greedy and exact optimizers, random search, legacy threshold Grover, and L2/L6 neural models. Cross-encoders on dense fifteen and dense sixty-four use different pools from L6-only on hybrid fifteen.

The seven expanded-pool policies comprise constrained GAS, constrained local-swap greedy, exact float, exact quantized, matched random search, global union-L6 ranking, and union-L6 with the protected prefix. Each constrained policy uses the actual baseline prefix for the seed and the same tail feasibility rules. The random refiner draws $\max(1, 2Q)$ uniform subsets in addition to its initial incumbent, with Q denoting the seed's recorded refinement phase count. That matches a declared objective-access allowance, not every operation of the compact simulator.

The exact controls independently enumerate their float or quantized objectives. The GAS controller receives neither their optimum nor the marked-state count. A comparison across pools describes complete policies; one among constrained refiners with identical features isolates their search outcomes for the same objective and admissible domain. These comparisons answer different questions.

D. METRICS, INFERENCE, AND HARDWARE

We use nDCG@5, Recall@8, and the redundancy measure in (2) as principal metrics; nDCG@8 shows effects at ranks six through eight. Discounted gains use linear judgment values and logarithmic rank discount [34], and saved rankings are checked with NIST's trec_eval [35]. For a five-seed policy, we first average within each query and then across queries. Seeds do not create independent relevance judgments

TABLE 4. Reported full-test policies. SciFact/NFCorpus include refinement; FiQA uses the earlier base. All quantities within a row come from the same recorded policy.

Dataset	nDCG@5	nDCG@8	Recall@8	Redundancy ↓
SciFact	0.687833	0.698173	0.801111	0.433636
NFCorpus	0.368865	0.346506	0.152997	0.431771
FiQA	0.357490	0.376939	0.434063	0.531244

for a query.

For protected-run inference, the recorded procedure uses 10,000 paired bootstrap resamples with seed zero and 10,000 paired sign-randomization draws with seed one. Differences no larger than 10^{-12} are ties. Holm correction covers nine comparators by four metrics per protected dataset. The earlier FiQA nDCG audit covers six declared comparators and removes numerical averaging artifacts before testing. We retain those families and their pointwise intervals rather than choosing a new family to change apparent significance.

The recorded host runs Linux and Python 3.12.3 on an Intel Core i7-1355U with 12 logical CPUs and approximately 15.3 GiB of RAM. Compute libraries use two threads. Recorded software versions include NumPy 1.26.4, SciPy 1.13.1, CPU PyTorch 2.12.0, Transformers 4.57.6, Sentence Transformers 3.4.1, Qiskit 0.46.0, and Aer 0.13.3 [52]. Online latency excludes model loading, corpus/cache restoration, and artificial-query warm-up. Session and host-load records are available, but the experiments were not run in a dedicated latency laboratory.

Online time includes the applicable retrieval, lexical, neural, and selection stages; reused upstream work is charged to each policy that requires it. Base and refinement times are summed for each seeded request before combined selector percentiles are calculated. Request P95 is not divided by the number of seeds. Section VII decomposes mean cost additively, without treating sums of component percentiles as pipeline percentiles.

VI. RECORDED RETRIEVAL RESULTS

A. EFFECTIVENESS UNDER THE DECLARED CONFIGURATIONS

Table 4 reports the selected full-test policies from the completed experiments; the journal analysis collects no new experiments. The nDCG@5 values are 0.6878 for SciFact, 0.3689 for NFCorpus, and 0.3575 for FiQA. For the first two, prefix preservation fixes nDCG@5 at the matched-base value. FiQA uses the earlier fifteen-candidate base selector, so its row represents a different configuration.

Relative to the matched base, SciFact refinement reduces mean redundancy by 3.21% and changes Recall@8 by -0.93% relative. NFCorpus has a 2.63% redundancy reduction and a +1.95% relative recall change. Both preserve the prefix metric, but recall moves in opposite directions. The refinement is not a general recall improvement.

The FiQA base reaches Recall@8 of 0.4341 and redundancy of 0.5312; its nDCG@5 and recall match the same-shortlist L6-only control. A later, separately completed 648-query refinement follow-up preserved nDCG@5 and reduced redundancy by 8.23%, but lost 2.60% relative recall. It failed the frozen 98% recall-retention gate and was not retuned to pass. The failed result is disclosed separately rather than mixed into the earlier base row.

B. CLASSICAL AND NEURAL COMPARISONS

The controls in Table 5 do not all see the same candidates. L6-only uses hybrid fifteen, while the dense-fifteen neural control receives the fifteen highest dense-scoring parents. Differences between them therefore include candidate construction. Dense-sixty-four reranking gives the relevance model more candidates and incurs the corresponding inference cost.

For SciFact and NFCorpus, matched greedy and exact-float refiners attain the same aggregate nDCG@5, nDCG@8, Recall@8, and redundancy as constrained GAS. The reported benefit can therefore be obtained by several optimizers under these candidates and constraints. In these tests, the measured aggregate quality is also reached without the Grover search mechanism.

Table 6 helps separate expansion from the choices made afterward. Global union-L6 can reorder the entire output. Protected-prefix union-L6 retains the head and selects the tail by neural relevance, while constrained optimizers also minimize pair redundancy under the score thresholds. Calling all of these policies ‘‘reranking’’ would obscure which part changed.

Figure 2 shows the resulting operating points. Some inexpensive or less relevance-focused policies have lower redundancy and lower recall. These are observed points, not a tuned Pareto frontier or proof of dominance over all settings. In particular, the zero-diversity MMR point does not represent the full range of MMR choices.

C. PAIRED UNCERTAINTY AND INTERPRETATION

Table 7 gives paired differences against the base for the protected datasets. SciFact's mean recall difference is negative, and its interval spans loss and gain. NFCorpus has a positive pointwise recall interval, but the recorded raw probability does not lead to rejection under the full Holm family. The interval and familywise decision address different questions.

The empirical 98% mean-recall gate is descriptive; it is not a statistical noninferiority test. Nor does an interval containing zero establish equivalence. Stronger inference would require a justified noninferiority margin, prospective analysis, and appropriate control of the data's role in development. We report the recorded gate outcomes and paired comparisons within those limits.

The FiQA audit also exposed a numerical issue. Averaging five identical seeded scores can differ from a deterministic control by approximately one unit in the last place; a significance test should not treat that artifact as a retrieval gain. The recorded 10^{-12} tie policy addresses it without changing rankings or reported raw means.

TABLE 5. Recorded full-test comparison. n@5 is nDCG@5, R@8 is Recall@8, and Red. is redundancy. Matched search controls use the refined domain on SciFact/NFCorpus and the base domain on FiQA. Additional saved methods remain in the complete CSV export.

	SciFact			NFCorpus			FiQA		
Policy	n@5	R@8	Red.	n@5	R@8	Red.	n@5	R@8	Red.
Dense cosine	0.6361	0.8111	0.4756	0.3181	0.1335	0.4398	0.3470	0.4153	0.5363
Dense/BM25 RRF	0.6677	0.8149	0.4727	0.3491	0.1394	0.4394	0.3509	0.4262	0.5327
Legacy threshold Grover	0.6364	0.8111	0.4756	0.3181	0.1334	0.4398	0.3458	0.4150	0.5363
Cosine-guided GAS	0.6311	0.7742	0.4490	0.3160	0.1235	0.3996	0.3452	0.4093	0.5215
L6 relevance only	0.6878	0.8086	0.4480	0.3689	0.1500	0.4447	0.3575	0.4341	0.5313
L6 MMR	0.6878	0.8086	0.4480	0.3689	0.1500	0.4447	0.3575	0.4341	0.5313
Base L6-guided GAS	0.6878	0.8086	0.4480	0.3689	0.1501	0.4434	0.3575	0.4341	0.5312
Matched greedy	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4341	0.5311
Matched exact float	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4343	0.5309
Matched exact integer	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4341	0.5311
Matched random	0.6878	0.8011	0.4338	0.3689	0.1531	0.4320	0.3575	0.4341	0.5313
L2, dense 15	0.6431	0.8049	0.4642	0.3509	0.1396	0.4486	0.3319	0.4217	0.5295
L6, dense 15	0.6790	0.8104	0.4655	0.3621	0.1423	0.4537	0.3575	0.4341	0.5313
L2, dense 64	0.6321	0.7916	0.4329	0.3481	0.1438	0.4468	0.2973	0.3918	0.4962
L6, dense 64	0.6815	0.8087	0.4379	0.3670	0.1437	0.4586	0.3535	0.4160	0.5033
Reported GAS policy	0.6878	0.8011	0.4336	0.3689	0.1530	0.4318	0.3575	0.4341	0.5312

TABLE 6. Expanded-pool controls on the two protected-refinement test runs. All four metrics are shown to expose tail effects hidden by a fixed top five. FiQA has no corresponding expanded-pool headline in the selected source mapping.

	SciFact				NFCorpus			
Policy	n@5	n@8	R@8	Red.	n@5	n@8	R@8	Red.
Union L6, global order	0.6860	0.6972	0.8037	0.4475	0.3688	0.3472	0.1522	0.4477
Union L6, protected prefix	0.6878	0.6993	0.8037	0.4474	0.3689	0.3466	0.1523	0.4474
Constrained GAS	0.6878	0.6982	0.8011	0.4336	0.3689	0.3465	0.1530	0.4318
Constrained greedy	0.6878	0.6982	0.8011	0.4336	0.3689	0.3465	0.1530	0.4318
Constrained exact float	0.6878	0.6982	0.8011	0.4336	0.3689	0.3465	0.1530	0.4318
Constrained exact integer	0.6878	0.6982	0.8011	0.4336	0.3689	0.3465	0.1530	0.4318
Constrained random	0.6878	0.6982	0.8011	0.4338	0.3689	0.3468	0.1531	0.4320

TABLE 7. Recorded paired inference: reported refinement minus base on SciFact/NFCorpus. †FiQA is its earlier base minus L6-only for the independently audited nDCG@5 comparison. Intervals are pointwise, not simultaneous; Holm decisions use each original full family.

Dataset	Metric	Mean difference	Paired 95% CI	Raw p	Holm reject
SciFact	nDCG@5	+0.00000	[+0.00000, +0.00000]	1.00000	No
SciFact	nDCG@8	-0.00268	[-0.00811, +0.00231]	0.30427	No
SciFact	Recall@8	-0.00750	[-0.02417, +0.00750]	0.45645	No
SciFact	Redundancy	-0.01439	[-0.01680, -0.01216]	0.00010	Yes
NFCorpus	nDCG@5	+0.00000	[+0.00000, +0.00000]	1.00000	No
NFCorpus	nDCG@8	+0.00094	[-0.00248, +0.00445]	0.58974	No
NFCorpus	Recall@8	+0.00293	[+0.00038, +0.00622]	0.03690	No
NFCorpus	Redundancy	-0.01165	[-0.01405, -0.00938]	0.00010	Yes
FiQA	nDCG@5†	0.00000	[0.00000, 0.00000]	1.00000	No

D. LATENCY AND OPTIMIZATION OUTCOMES

Complete online median and P95 times appear in Table 8. Refinement itself is small, but expanded-pool inference makes the full two-stage pipeline slower than the matched base. Dense-sixty-four rerankers differ in candidate opportunity as well as cost, so the comparison does not support a universal speed claim.

Table 9 separates search work from retrieval quality. An exact objective hit is measured against the corresponding exact float-redundancy optimum for refinement and the exact integer optimum for the base. Neither certifies maximum judged recall or nDCG. Budget exhaustion, a trivial basis, and an unchanged incumbent remain in the request denominator.

The counts describe this implementation rather than physical acceleration. Because the CPU simulator prepares the full objective array, classical methods can inspect it directly. The objective assumptions and measured cost boundaries remain part of the comparison.

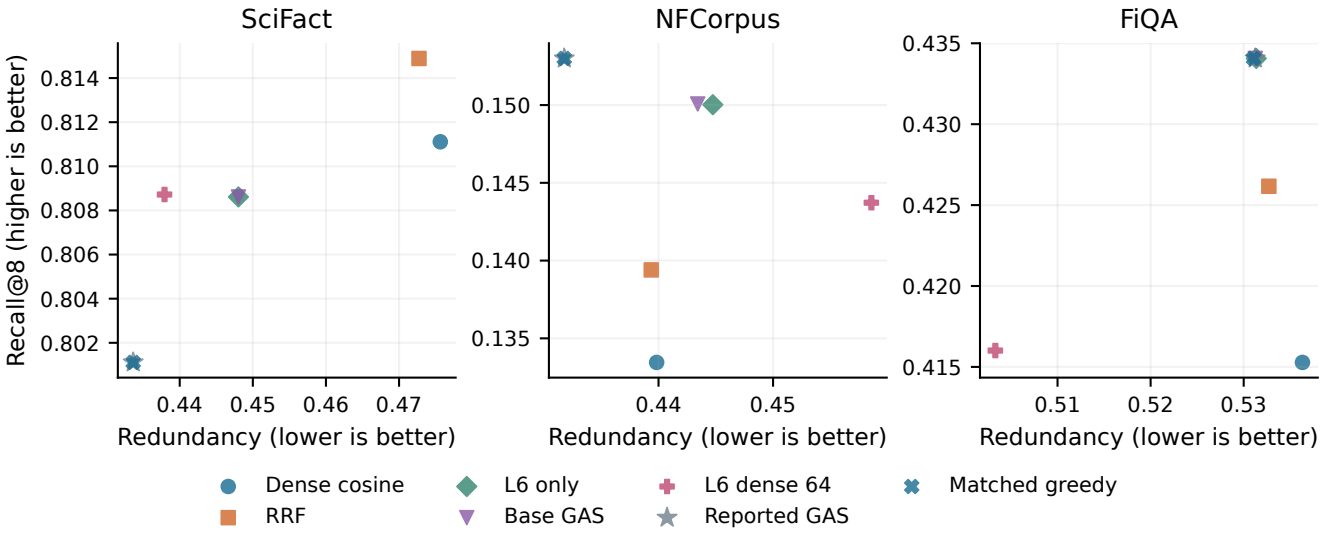


FIGURE 2. Observed Recall@8–redundancy operating points from the saved runs. Coincident points can arise when policies return the same effective outputs. Candidate pools differ as specified in the protocol; the graph is descriptive and does not imply matched causal comparisons across every point.

TABLE 8. Composed online latency, milliseconds per seeded request. Values include all applicable upstream stages. P50/P95 are computed over recorded request totals; restoration, model loading, and warm-up are separate.

	SciFact		NFCorpus		FiQA	
Policy	P50	P95	P50	P95	P50	P95
Dense cosine	33.3	48.3	24.5	32.9	79.9	98.5
Dense/BM25 RRF	61.6	94.9	31.0	44.5	225.3	328.3
Legacy threshold Grover	38.2	313.6	25.2	281.6	84.3	231.9
Cosine-guided GAS	42.3	56.8	33.7	42.0	84.1	102.5
L6 relevance only	955.3	1085.4	851.9	901.4	650.0	821.1
L6 MMR	955.5	1085.7	852.1	901.6	650.1	821.3
Base L6-guided GAS	963.2	1093.7	859.8	909.1	653.8	826.7
Matched greedy	1405.4	1670.7	1145.5	1479.6	650.0	821.2
Matched exact float	1405.2	1670.6	1145.3	1479.3	651.2	822.5
Matched exact integer	1405.2	1670.6	1145.3	1479.3	651.1	822.4
Matched random	1405.2	1670.6	1145.4	1479.3	651.9	824.8
L2, dense 15	348.0	403.0	310.5	331.5	238.9	294.6
L6, dense 15	915.8	1029.2	832.5	882.5	539.6	618.2
L2, dense 64	1374.7	1562.5	1242.5	1319.6	763.7	912.2
L6, dense 64	3799.0	4322.2	3460.8	3630.9	2037.9	2437.9
Reported GAS policy	1406.5	1672.3	1146.8	1481.3	653.8	826.7

VII. JOURNAL EXTENSION: COVERAGE, FAILURE, AND COST DIAGNOSTICS

A. CANDIDATE COVERAGE SEPARATES RETRIEVAL FROM SELECTION

Saved feature and ranking records let us reconstruct candidate membership and evaluate (4) for dense sixty-four, hybrid fifteen, and the expanded union where available. We check pool sizes, subset inclusion, and parent IDs independently of historical field names before computing coverage. No new rankings are created.

Table 10 and Fig. 3 show how much candidate opportunity differs by collection. SciFact's ceiling is 0.8569 for hybrid fifteen and 0.8903 for the expanded union. The corresponding NFCorpus ceilings are 0.1725 and 0.1942. FiQA's base shortlist has a ceiling of 0.4824, compared with 0.6483 for dense sixty-four. These are optimistic opportunities under the judgments, not predictions of a learned reranker's performance.

For 40 SciFact, 91 NFCorpus, and 203 FiQA queries, the hybrid shortlist contains no positively judged parent. An eight-of-fifteen selector cannot improve judged recall on those queries without different candidates. Expansion recovers some exclusions in the protected datasets, but stays inside dense sixty-four and cannot recover a corpus document missing from that set.

The number of judged-relevant documents also changes the scale of Recall@8. Even eight perfect outputs recover only a fraction of a large relevant set. NFCorpus contains queries with substantially larger relevant sets than the single-evidence cases common in SciFact; its lower recall reflects task structure as well as candidate opportunity and rank-

TABLE 9. Recorded search work and combined selector latency in milliseconds. The first phase column counts inherited base work for refined policies; it is zero for base-only FiQA. The next column counts the current stage. Optimum-hit definitions follow the text.

Dataset	Requests	Base phases	Refine/base phases	Optimum-hit rate	Selector P50	Selector P95
SciFact	1500	64.00	20.08	1.000	9.66	10.67
NFCorpus	1615	63.91	17.82	1.000	9.49	11.02
FiQA	3240	0.00	64.00	0.994	3.95	4.54

TABLE 10. New secondary analysis of saved candidate pools. Judged coverage includes every positively judged parent in the pool. The Recall@8 ceiling additionally enforces the eight-output budget and relaxes relevance-score and prefix constraints. Values are query means.

Dataset	Pool	Mean size	Judged coverage	Recall@8 ceiling
SciFact	Dense 64	64.00	0.9243	0.9243
SciFact	Hybrid 15	15.00	0.8569	0.8569
SciFact	Expanded union	22.30	0.8903	0.8903
NFCorpus	Dense 64	64.00	0.2666	0.2375
NFCorpus	Hybrid 15	15.00	0.1780	0.1725
NFCorpus	Expanded union	20.29	0.2028	0.1942
FiQA	Dense 64	64.00	0.6489	0.6483
FiQA	Hybrid 15	15.00	0.4824	0.4824

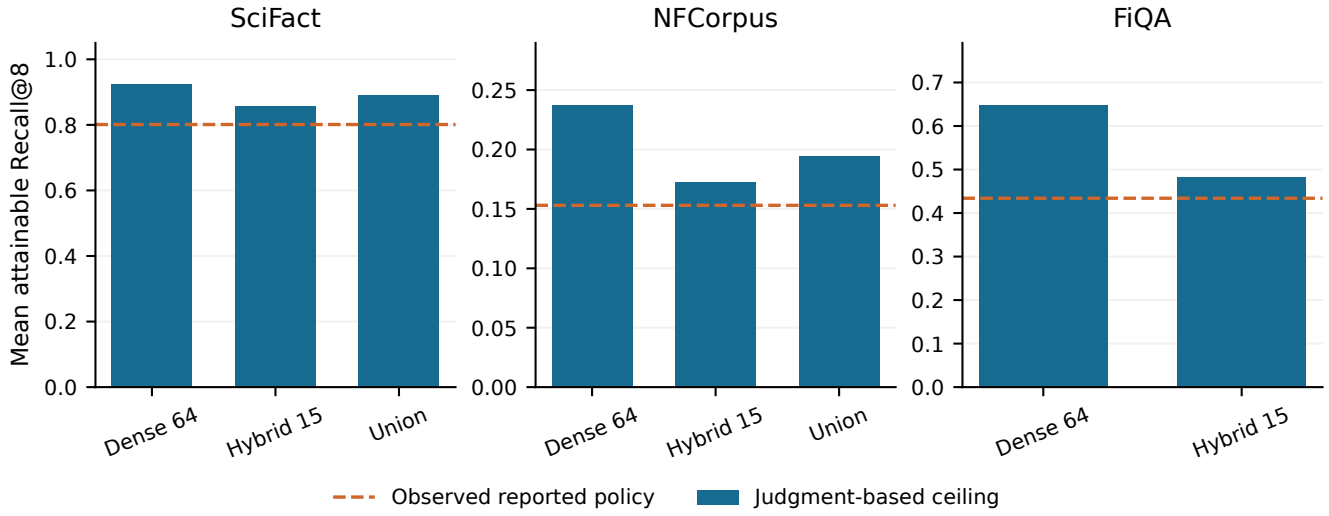


FIGURE 3. New judgment-based Recall@8 ceilings for the saved candidate pools, compared with the observed reported-policy recall. The dashed line is the same policy within each dataset panel. Expanded-union ceilings relax the protected-prefix and tail-score constraints and therefore do not measure pure search inefficiency.

ing performance. Comparing policies on the same queries, against the relevant candidate ceiling, helps separate these factors.

The union widens the relaxed opportunity set, but the actual task commits five positions to the original head and requires replacements to satisfy neural-score dominance. Redundancy remains a proxy for complementary evidence. The gap to the union ceiling therefore reflects constraints and model errors as well as the finite search budget.

B. MEAN IMPROVEMENTS COEXIST WITH INDIVIDUAL LOSSES

Table 11 counts changes from the reference policy after seed averaging within each query. Against base GAS, SciFact has 2 recall gains, 5 losses, and 293 ties; NFCorpus has 15 gains, 13 losses, and 295 ties. The favorable NFCorpus mean thus includes queries that lose recall. FiQA has no recall changes beyond the tie tolerance against L6-only, although some memberships and redundancy values differ.

Redundancy falls for 158 SciFact queries and 135 NFCorpus queries, and rises for none, consistent with Proposition 2. The paired changes in Fig. 4 show many zeros and a small subset with recall differences. A mean does not show how those outcomes are distributed.

The following extreme cases were chosen deterministically to illustrate the risk, not to represent the full test set. SciFact query 384 has one positively judged relevant parent. Base recall is 1 and refined recall is 0, although the candidate-pool

TABLE 11. New query-level outcome and seed-stability analysis. References are base GAS for SciFact/NFCorpus and L6-only for FiQA. Counts use a 10^{-12} numerical tie tolerance. Seed-varying ranks count queries with more than one final ordered list among the five saved seeds.

Dataset	Recall wins	Ties	Losses	Lower redundancy	Seed-varying ranks	Seed-varying recall
SciFact	2	293	5	158	0	0
NFCorpus	15	295	13	135	29	3
FiQA	0	648	0	6	7	0

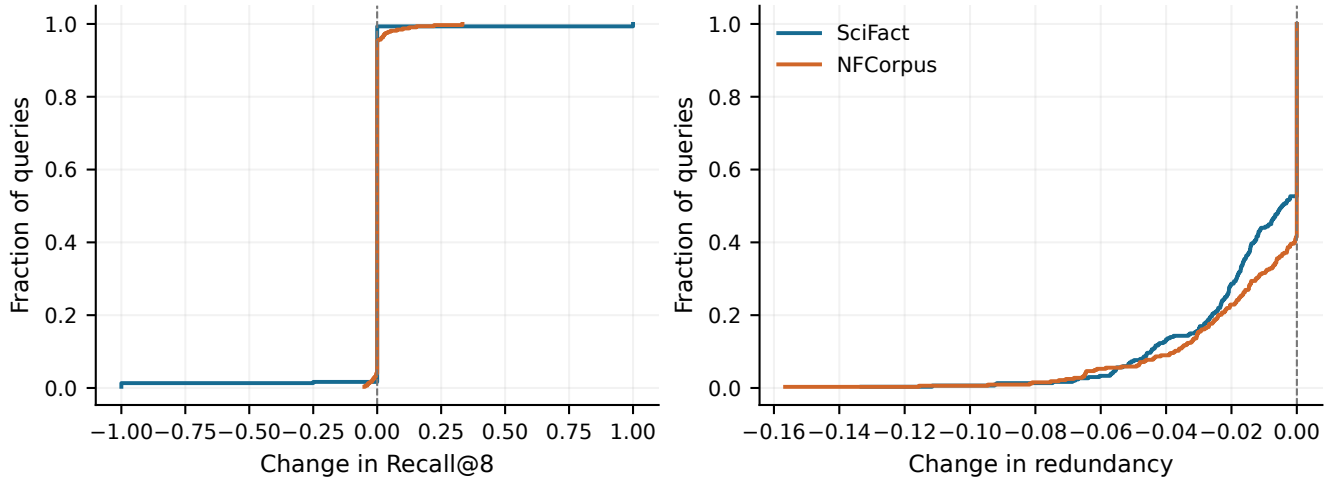


FIGURE 4. New empirical cumulative distributions of query-level refinement-minus-base differences. Recall changes can be positive or negative. Nonpositive redundancy changes follow the refinement acceptance rule. Seeds are averaged within queries before these distributions are constructed.

ceiling is 1 and the top five are unchanged. The only relevant output was lost below the protected head. Redundancy fell by approximately 0.04373 even as judged retrieval worsened.

For NFCorpus query PLAIN-2570, the judgments contain 40 relevant parents. Base Recall@8 is 0.075 and refined Recall@8 is 0.025, or three versus one positively judged output. Mean redundancy falls by approximately 0.09454, while the relaxed Recall@8 ceiling is 0.125 for both the original and expanded pools. Candidate opportunity alone does not explain the loss; admissibility, relevance estimates, and the objective also matter. The fixed retrospective selection rule used lowest paired recall change, then largest redundancy decrease, then query ID.

Protecting a top-five metric can serve an application that values that prefix, but it does not establish that the remaining evidence can be replaced safely. A mean-retention threshold may hide a severe loss on an individual query. Recall distributions, the number of harmed queries, and their application-specific costs should accompany the mean.

C. SEED STABILITY IS A PROPERTY OF OUTPUTS, NOT INDEPENDENT EVIDENCE

Across seeds, final rankings differ on 0 SciFact, 29 NFCorpus, and 7 FiQA queries. Recall varies on 0, 3, and 0 queries, respectively. Different sampled trajectories can lead to the same final output; the five-seed average does not by itself describe that distinction.

Stable outputs do not create independent queries, and different tail documents can share a judged status without changing recall. On NFCorpus, more queries have varying ordered lists than varying recall. The complete seed export retains scores and search statuses, allowing this difference to be inspected directly rather than inferred from aggregate variance.

When stochastic search repeatedly reaches an output also found by matched greedy and exact methods, the result does not identify randomness or amplitude amplification as the unique cause of the effectiveness gain. It does describe this implementation's stability under the recorded budgets and seeds.

D. ADDITIVE COST ATTRIBUTION AND THE SCOPE FOR SELECTOR ACCELERATION

For each primary seeded request, the journal exporter reconstructs

$$T_{\text{online}} = T_{\text{upstream}} + T_{\text{L6,original}} + T_{\text{L6,extra}} + T_{\text{base}} + T_{\text{refine}}. \tag{21}$$

Upstream time covers query encoding, dense search, representative features, and applicable lexical/hybrid/union processing. Extra inference and refinement are zero for the FiQA base. We check each sum against the saved request total before taking means. Composed stage measurements are additive; separately measured percentiles are not necessarily so.

In Fig. 5, original L6 inference accounts for 64.0%, 71.3%, and 64.8% of the respective primary mean online times.

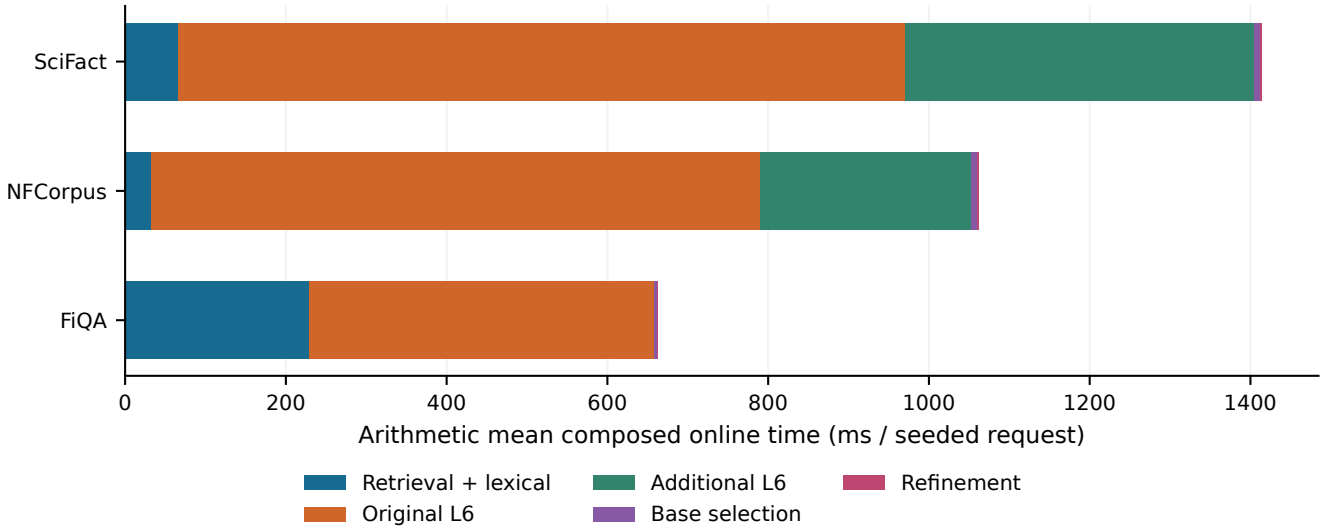


FIGURE 5. **New decomposition of arithmetic mean online time using additive per-request components. Small selector segments are retained. The plot uses means for additivity; Table 8 separately reports request-level medians and P95 values. Differences between collection sessions are not controlled cross-session speed experiments.**

Additional L6 scoring further dominates the refined pipelines. Base-plus-refinement selection accounts for 0.67%, 0.88%, and 0.61%, respectively. Even an idealized removal of all measured selector time would leave most of the composed request cost.

For $f = \overline{T}_{\text{selector}}/\overline{T}_{\text{online}}$, a hypothetical zero-time selector with unchanged upstream stages could reduce the mean by fraction f. The idealized speedup would be $1/(1-f)$. This bound accounts for the same staged computation; it is not a hardware prediction. A practical quantum implementation would still need state preparation, data access, reversible arithmetic, and measurement. Within the measured cost boundary, reduced selection phase counts alone give little support to an end-to-end acceleration claim.

VIII. RECORDED RESOURCE STUDY AND PRECISION SENSITIVITY

A. COMPACT SIMULATION AND NATIVE CIRCUITS SOLVE DIFFERENT ACCOUNTING TASKS

The compact selector stores amplitudes only for subsets of the required size. Its complex128 array occupies $16F$ bytes, with $F = \binom{15}{8}$ for the base and $F = \binom{M}{3}$ for refinement. Stored coefficients, energies, candidate features, and process memory are additional. A full ν-qubit statevector would require $16\,2^{\nu}$ bytes for amplitudes alone. Neither expression measures total system memory.

Compact simulation enumerates basis energies before amplification. Those objective evaluations are separate from incumbent checks, measured trials, and logical phase calls. The recorded convention assigns two reversible objective calls to a phase operation for compute and uncompute. Actual circuit execution would add constraint evaluation, state preparation, reflection, and physical data access; logical queries alone leave those costs out.

The native study uses 32 saved training queries per collection from the confirmation samples. It evaluates five compact-search seeds, tail pools of 3, 5, 8, 12, 16, or 20 candidates, and pair precisions of 4, 8, 12, or 16 bits. Of the 178 completed tasks, 80 are native synthesis cells. When a requested pool size is unavailable for a real query, it is not silently replaced by a smaller pool. Artificial stress instances at the specified larger sizes are labeled separately.

This study was completed before the manuscript and adds neither relevance judgments nor another full-test evaluation. Each controlled subpool retains the original tail. Native representatives, when available, are the first two eligible frozen seed-zero queries for a dataset/pool; they stay fixed across precisions. Pool sizes can still involve different eligible queries, so the growth curves describe measured cells rather than a random-population scaling law.

B. REVERSIBLE CONSTRAINT AND OBJECTIVE REGISTERS

Writing $x_i \in \{0,1\}$ for a remaining tail candidate, the integer refinement objective has the polynomial form

$$E(x) = C_H + \sum_i \alpha_i x_i + \sum_{i<j} \alpha_{ij} x_i x_j, \tag{22}$$

Here C_H contains fixed head–head terms, α_i sums rounded head–candidate terms, and α_{ij} is a rounded candidate-pair term. The oracle evaluates signed improvement from the incumbent and marks only states of cardinality three that satisfy all three threshold-count conditions in (14).

The register allocation comprises M decision qubits, w signed-value qubits, $c = \lceil \log_2(M+1) \rceil$ counter qubits, $c-1$ comparator workspace qubits, and four constraint flags. Thus,

$$\nu = M + w + 2c + 3. \tag{23}$$

TABLE 12. Recorded native-oracle synthesis summary. Slash-separated entries compare 4/16-bit coefficients; full four-precision measurements are retained in the accompanying native CSV. Width includes workspace. Depth and CX values are maxima over the stated bounded sample; stress rows are artificial.

Tail	Source	Qubits	Depth	CX gates
3	real	19/31	1049/1865	992/2408
5	real	23/35	3891/5955	3256/6160
8	real	29/41	15067/20722	11684/18240
12	real	34/46	27570/41294	22068/36076
16	stress	39/51	61665/67687	43628/52212
20	stress	43/55	79769/96137	58220/75556

Signed width must accommodate objective differences on every bit string, including those with invalid cardinality. Choosing w from valid three-element subsets alone could permit overflow before invalid states are rejected. The implementation bounds the full polynomial domain and counts every workspace register in the reported total.

The oracle computes the objective and constraints, applies the phase conditionally, then uncomputes. Counter and comparison workspace can be reused across threshold checks, but their flags must be retained until phase marking. This register structure determines the logical-width formula; the compact classical basis dimension is not an interchangeable qubit count.

The measured cells are summarized in Table 12; Figure 6 shows all four precision levels and separates real samples from artificial stress cases. Synthesis uses the unrestricted u, cx basis, optimization level one, and transpiler seed zero. The results are logical resources without coupling-map routing, device calibration, or error correction, so the gate count is not an execution cost for a particular processor.

C. PRECISION AND SEARCH ERROR

The coefficient bound in (16) shrinks with b, but the GAS output may still be farther from the exact float optimum when a finite search budget misses the exact integer optimum. Figure 7 separates the empirical search outcome from coefficient approximation. The saved float gap is output redundancy minus the exact admissible float optimum; the quantized gap uses the exact encoded optimum.

More precision can separate objective values at the cost of wider signed arithmetic and more circuit work. A stochastic output with a fixed budget need not improve monotonically as precision increases. Nor does a smaller optimization gap imply better judged relevance: the objective still penalizes pair similarity under estimated-score constraints. We report retrieval metrics, exact objective controls, and resources together for that reason.

D. VALIDATION COVERAGE AND EXPLICIT LIMITS

Complete-iteration synthesis includes generic fixed-weight preparation and reflection for at most eight decision qubits. Native validation simulation covers at most five decisions. Six complete iterations passed phase-sensitive validation; a conservative memory guard refused 42 other simulation attempts. Those refusals are limits, not successful executions of larger circuits. Separate small two-bit toy checks exercise nontrivial five-decision marking and reflection.

Native validation uses a noiseless matrix-product-state simulator with zero configurable truncation threshold, maximum bond dimension 128, and probability tolerance 10^{-8}. An independently constructed inverse circuit for the expected state is followed by local Z expectations, which bound state mismatch and workspace leakage. Each circuit worker has 120-second and 2 GiB limits. Larger native simulations, a noise sweep, hardware routing, and physical execution were not performed.

Construction, transpilation, native validation, compact selection, allocation profiling, and full-pipeline timing cover different work. Per-method allocation profiling is a separate replay and is excluded from selector timing; the fresh worker's process RSS covers its complete grid. These boundaries prevent the replay from inflating one time while an expensive prerequisite is omitted from another.

IX. DISCUSSION, LIMITATIONS, AND REPRODUCIBILITY

A. WHAT THE COMBINED EVIDENCE ESTABLISHES

The protected head explains part of the outcome by construction: a fixed prefix preserves the corresponding truncated ranking metric. What happens in the tail is empirical. Pair similarity falls on many queries, but relevance can change when candidate availability or neural scores are unfavorable. The coverage and loss analyses show why the guarantee and the losses can coexist.

The matched greedy and exact selectors reach the same reported aggregate quality as GAS. They share its candidate pool, objective, and constraints, so the gain over base selection cannot be attributed to amplitude amplification alone. The constrained domains are small enough to make exact controls practical, allowing that distinction to be tested.

The request-cost breakdown places a practical limit on the efficiency claim. Selector arithmetic occupies only a small part of the recorded pipeline; neural relevance inference takes a substantial share. Fewer phase queries alone cannot be assumed to produce a comparable end-to-end latency reduction when candidate scoring and upstream work remain. This conclusion concerns the recorded pipeline and host; a substantially different access model would need its own measurements.

B. CONSTRUCT AND MEASUREMENT VALIDITY

The redundancy measure averages clipped embedding similarities. It does not directly measure repeated facts, contradiction, complementary evidence, or answer faithfulness. Necessary evidence can be correlated and incur a high penalty; irrelevant text can be dissimilar and incur a low one. An application-level study is needed to determine whether reducing this proxy helps the intended task.

Judgments concern parent documents, but relevance inference sees a representative passage. Supporting informa-

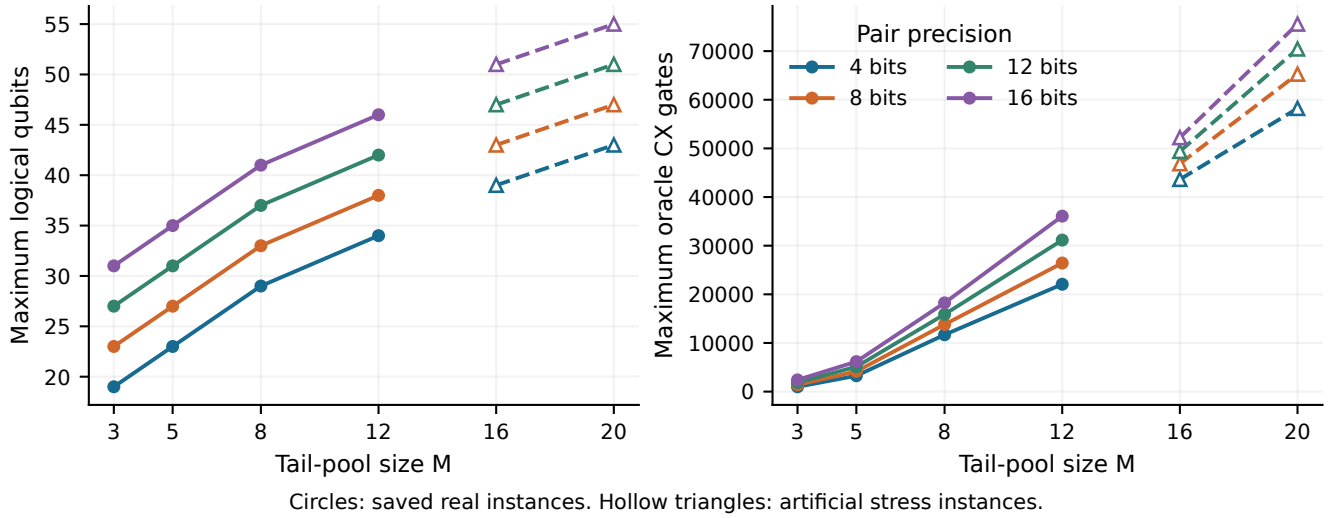


FIGURE 6. **All recorded precision levels for native logical width and oracle CX count. Real-instance points summarize the available fixed representatives; hollow triangles are artificial stress cells. Connecting lines aid inspection and are not fitted scaling models.**

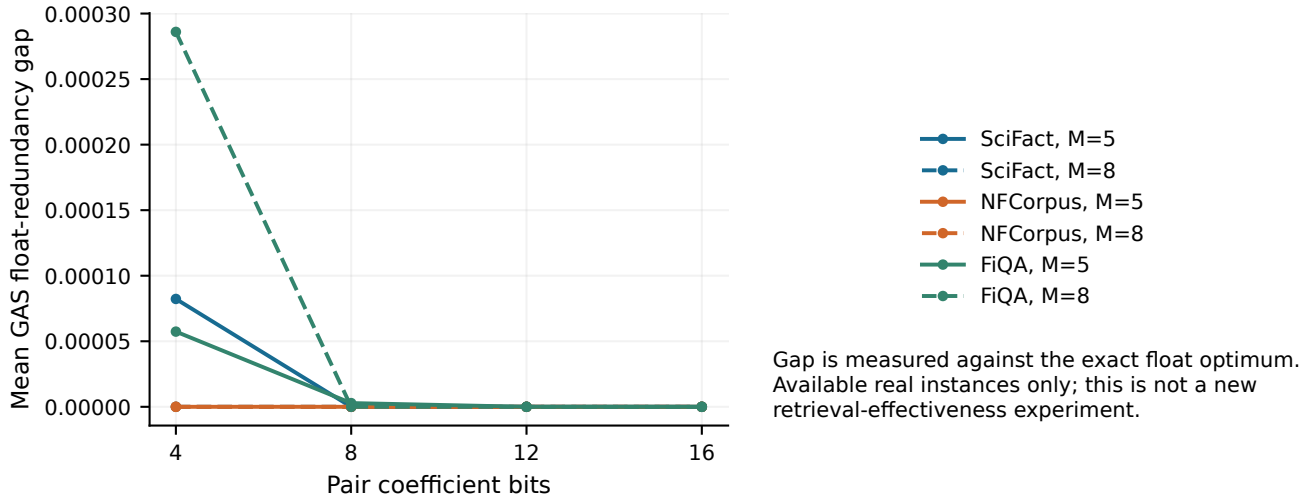


FIGURE 7. **Recorded GAS float-redundancy gaps across pair precision for available real five- and eight-candidate subpools. Values are means over the study's fixed available requests, not new retrieval runs. Search error can dominate coefficient error; the exact-optimizer regret bound does not bound these GAS curves.**

tion may lie outside the parent's highest-cosine chunk, and the passage score may not describe the whole document. Dense top-64 filtering, hybrid top-fifteen filtering, and score-dominance constraints can each remove useful candidates. The judgment-based ceilings expose some losses; incomplete judgments mean they are not ceilings on all human relevance.

Preserving nDCG@5 does not settle the quality of the full set. Recall@8 and nDCG@8 may change, individual queries may lose substantial recall, and a downstream consumer may use the lower-ranked tail. The descriptive mean-retention gate also provides less evidence than a prospective statistical non-inferiority design. These failures qualify the reported gains.

C. INTERNAL AND EXTERNAL VALIDITY

Because the official tests were reused during development, their earlier observation limits a confirmatory interpretation even though the recorded follow-up settings were not retuned on those outcomes. We report the failed NFCorpus training confirmation and the decision to continue. The failed FiQA refinement follow-up likewise prevents a claim of cross-domain success for the protected policy based on the earlier base row. The diagnostics and extreme examples are retrospective.

SciFact, NFCorpus, and FiQA cover different tasks, but not every domain, language, query distribution, or private collection. The measured L2/L6 controls do not exhaust modern rerankers. Training-data overlap and domain mismatch may

affect the comparison, and a model-family name does not establish exact training/test disjointness. Prospective benchmarks and stronger rerankers are needed to assess generalization.

The latency measurements come from separate sessions on a shared host. The composition model charges applicable upstream costs to each policy but does not measure concurrency or production throughput. Warm-up and restoration are reported separately. Monetary cost, energy, quality of service, and physical quantum runtime were not measured. Translating unrestricted logical gate counts to device performance would require assumptions about routing, errors, and fault tolerance.

D. ARTIFACT AND NUMERICAL REPRODUCIBILITY

The journal artifact provides the LaTeX source, tables, figures, bibliography, analysis code, query diagnostics, seed metrics, and request timing components. Source mappings identify completed runs by manifest and evaluation hashes. Export reads those runs without collecting data, selecting documents, or performing neural inference; the journal scripts write within their own artifact directory.

Verification checks test-query, method, and seed coverage; source, model, and data identities; original logits; output cardinality; protected prefixes; and recorded feasibility. It independently recomputes 119,124 query–method–metric values from saved rankings and retains the experiment-bound NIST metric checks. Coverage diagnostics check the sizes and containment of dense, hybrid, and union candidate sets. A separate request-level check verifies the additive latency identity before the stacked mean figure is drawn.

The source collections and model records are public and cited, while the saved experimental artifacts remain local inputs without a public archival DOI. The journal's evidence exports make the present claims inspectable. Depositing the complete experiments would further support independent reproduction of the underlying runs.

E. FUTURE RESEARCH

The refined policy needs a prospective test on an untouched domain with relevance and loss criteria specified in advance. Such a study should retain failed outcomes, compare modern relevance models, and examine whether a full MMR tradeoff curve or another subset-diversity model offers a preferable operating point. A new generated-answer evaluation would be needed to connect changes in evidence sets to factuality, completeness, and answer usefulness.

Quantum implementation calls for further work on structured fixed-weight preparation, reversible coefficient access, hardware routing, noise, and fault-tolerant resource estimates. Comparisons must include classical algorithms solving the same admissible objective and the complete neural retrieval pipeline. Lower logical phase counts alone do not resolve the cost bottlenecks measured here.

X. CONCLUSION

We evaluated Grover-inspired evidence selection under explicit relevance, candidate, and resource constraints. The completed three-collection evidence shows that a protected five-document prefix can accompany both lower pairwise embedding similarity and losses in judged recall. The journal extension examines candidate-pool ceilings, individual-query outcomes, seed stability, and additive request costs without a new benchmark or a retuned policy. Matched classical optimizers reach the same reported aggregate quality on the protected tests, while original and additional cross-encoder inference dominate composed online costs. The separately completed reversible-oracle study reports growing logical resources and limited simulation coverage; it does not establish physical-device acceleration. These findings bound the relevance–redundancy formulation and provide a reproducible account of its behavior. Claims about generalization, generated answers, or quantum advantage require prospective effectiveness tests and independently measured implementation costs. The guaranteed prefix, empirical tail quality, and negative outcomes must remain distinct in that assessment.

APPENDIX A DETAILED QUANTIZATION AND CONSTRAINT REASONING

For the base objective, express each rounded coefficient as its exact scaled value plus an error in $[-1/2, 1/2]$. For a fixed k-element set, the total absolute error is bounded by

$$\frac{1}{2}\left(k + \binom{k}{2}\right) = \frac{k(k+1)}{4}. \tag{24}$$

Dividing by $A_b k$ yields (11). Errors may cancel, so the actual error can be much smaller than this worst-case bound. The result concerns a fixed set, not the outputs of two different stochastic optimization trajectories.

For refinement, let $e_A(S) = E_r(S)/(28A) - \mathcal{R}(S)$. Equation (16) states $|e_A(S)| \le 1/(2A)$. Exact optimality in the encoded objective implies

$$\mathcal{R}(S_A) + e_A(S_A) \le \mathcal{R}(S_\star) + e_A(S_\star), \tag{25}$$

$$\mathcal{R}(S_A) - \mathcal{R}(S_\star) \le e_A(S_\star) - e_A(S_A) \le 1/A. \tag{26}$$

The admissible domain must be the same for both optima. The comparison does not cover a quantized optimizer using another pool, fixed head, or score constraint. A GAS output that misses the exact encoded optimum has an additional optimization gap.

The equivalence between scores and counts includes ties. For the ascending triple $y_1 \le y_2 \le y_3$, $y_j \ge \theta_j$ holds exactly when at least $4 - j$ values are at least θ_j. If $y_j < \theta_j$, the first j entries fail the threshold, leaving at most $3 - j$ that can meet it. The same ordering argument gives the reverse implication. Applying it to each threshold proves (14) without requiring distinct logits.

TABLE 13. Recorded legacy development thresholds. nDCG@5 comes from saved matched-sampling outcomes. No-marked percentages count development queries, with no new ranking or test tuning.

Dataset	Threshold	Dev. nDCG@5	No-marked queries (%)
SciFact	0.3	0.6627	0.0
SciFact	0.5	0.6579	6.8
SciFact	0.7	0.6656	67.1
NFCorpus	0.3	0.3010	0.6
NFCorpus	0.5	0.3011	31.8
NFCorpus	0.7	0.3026	87.0
FiQA	0.3	0.3514	0.0
FiQA	0.5	0.3515	1.8
FiQA	0.7	0.3516	69.4

APPENDIX B NATIVE VALIDATION BOUND AND REPORTING CONVENTIONS

After applying the independent expected-state inverse, the ideal joint decision/workspace state is all zero. If $p_j = (1 - \langle Z_j \rangle)/2$ denotes the probability that qubit j is one, a union bound gives

$$1 - |\langle 0 \cdots 0 | \psi \rangle|^2 \leq \sum_j p_j. \tag{27}$$

Restricting the sum to workspace qubits bounds the probability of dirty workspace as well. Native validation checks the bound and norm against 10^{-8} for the small instances actually simulated. The check is phase-sensitive for those circuits; it does not validate the 42 refused simulations or other unrun instances.

Table 13 retains the saved legacy threshold analysis. Its no-marked percentage is calculated over queries with recorded cosine candidates; seeds are not treated as independent events. Development outcomes come from saved matched-sampling rankings, not a new threshold sweep on test judgments. A lower threshold can reduce empty marked sets but also change the ranking policy, making fallback rate insufficient as an effectiveness criterion.

The complete exports include all twenty or twenty-seven measured methods, even where compact manuscript tables omit controls. The query diagnostic CSV identifies reference policies and candidate pools; the seed CSV retains request status and phase counts. Numerical ties consistently use a 10^{-12} tolerance. The native resource CSV distinguishes validated, memory-limited, unattempted, and synthetic cases. Thus a missing measurement, a retained incumbent, and a successful quantum simulation remain distinct events.

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REFERENCES

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474. [Online]. Available: https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html

[2] L. K. Grover, "A fast quantum mechanical algorithm for database search," in *Proceedings of the twenty-eighth annual ACM symposium on Theory of computing - STOC '96*, 1996, pp. 212–219, doi: 10.1145/237814.237866.

[3] G. Brassard, P. Høyer, M. Mosca, and A. Tapp, "Quantum amplitude amplification and estimation," in *Quantum Computation and Information*, ser. Contemporary Mathematics, vol. 305, 2002, pp. 53–74, doi: 10.1090/conm/305/05215.

[4] B. R. Gundapaneni, A. B. Syed, J. Borru, and N. S. Puppala, "Qubit Chat: A Controlled Evaluation of Grover-Inspired Reranking for Document Retrieval," 2026, accepted for the 6th International Conference on Evolutionary Computing and Mobile Sustainable Networks (ICECMSN); conference manuscript, awaiting publication.

[5] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009, doi: 10.1561/1500000019.

[6] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.

[7] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms condorcet and individual rank learning methods," in *Proceedings of the 32nd international ACM SIGIR conference on Research and development in information retrieval*, 2009, pp. 758–759, doi: 10.1145/1571941.1572114.

[8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.

[9] R. Nogueira and K. Cho, "Passage Re-ranking with BERT," arXiv:1901.04085, 2019. [Online]. Available: https://arxiv.org/abs/1901.04085

[10] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5776–5788. [Online]. Available: https://arxiv.org/abs/2002.10957

[11] P. Bajaj *et al.*, "MS MARCO: A Human Generated MAchine Reading COmprehension Dataset," arXiv:1611.09268, 2016. [Online]. Available: https://arxiv.org/abs/1611.09268

[12] Sentence Transformers, "all-MiniLM-L6-v2 Model Card," Hugging Face, accessed: Oct. 24, 2026. [Online]. Available: https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[13] ——, "MS MARCO MiniLM Cross-Encoder Model Cards," Hugging Face, l2 control: https://huggingface.co/cross-encoder/ms-marco-MiniLM-L2-v2. Accessed: Oct. 24, 2026. [Online]. Available: https://huggingface.co/cross-encoder/ms-marco-MiniLM-L6-v2

[14] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense Passage Retrieval for Open-Domain Question Answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781, doi: 10.18653/v1/2020.emnlp-main.550.

[15] O. Khattab and M. Zaharia, "ColBERT," in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020, pp. 39–48, doi: 10.1145/3397271.3401075.

[16] R. Nogueira, Z. Jiang, R. Pradeep, and J. Lin, "Document Ranking with a Pretrained Sequence-to-Sequence Model," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 708–718, doi: 10.18653/v1/2020.findings-emnlp.63.

[17] W. Sun, L. Yan, X. Ma, S. Wang, P. Ren, Z. Chen, D. Yin, and Z. Ren, "Is ChatGPT Good at Search? Investigating Large Language Models as Re-Ranking Agents," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 14 918–14 937, doi: 10.18653/v1/2023.emnlp-main.923.

[18] A. Abdallah, B. Piryani, J. Mozafari, M. Ali, and A. Jatowt, "How Good are LLM-based Rerankers? An Empirical Analysis of State-of-the-Art Reranking Models," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025, pp. 5693–5709, doi: 10.18653/v1/2025.findings-emnlp.305.

[19] H. Elkiran and J. Rasheed, "An Empirical Evaluation of Retrieval, Reranking, and Similarity for a Q&A-Based Retrieval Augmented Generation System," *IEEE Access*, vol. 14, pp. 26 053–26 066, 2026, doi: 10.1109/ACCESS.2026.3664852.

[20] J. Carbonell and J. Goldstein, "The use of MMR, diversity-based reranking for reordering documents and producing summaries," in *Proceedings of the 21st annual international ACM SIGIR conference on Research and development in information retrieval*, 1998, pp. 335–336, doi: 10.1145/290941.291025.

[21] A. Kulesza and B. Taskar, "Determinantal Point Processes for Machine Learning," *Foundations and Trends in Machine Learning*, vol. 5, no. 2-3, pp. 123–286, 2012, doi: 10.1561/2200000044.

[22] A. Gilliam, S. Woerner, and C. Gonciulea, "Grover Adaptive Search for Constrained Polynomial Binary Optimization," *Quantum*, vol. 5, p. 428, 2021, doi: 10.22331/q-2021-04-08-428.

[23] M. Boyer, G. Brassard, P. Høyer, and A. Tapp, "Tight Bounds on Quantum Searching," *Fortschritte der Physik*, vol. 46, no. 4–5, pp. 493–505, 1998. [Online]. Available: https://arxiv.org/abs/quant-ph/9605034

[24] C. Dürr and P. Høyer, "A Quantum Algorithm for Finding the Minimum," arXiv:quant-ph/9607014, 1996. [Online]. Available: https://arxiv.org/abs/quant-ph/9607014

[25] H. Ominato, T. Ohyama, and K. Yamaguchi, "Grover Adaptive Search With Fewer Queries," *IEEE Access*, vol. 12, pp. 74 619–74 632, 2024, doi: 10.1109/ACCESS.2024.3403200.

[26] M. Norimoto, T. Mikuriya, and N. Ishikawa, "Quantum Speedup for Multiuser Detection With Optimized Parameters in Grover Adaptive Search," *IEEE Access*, vol. 12, pp. 83 810–83 821, 2024, doi: 10.1109/ACCESS.2024.3413084.

[27] A. Bärtschi and S. Eidenbenz, "Deterministic Preparation of Dicke States," in *Fundamentals of Computation Theory*, 2019, pp. 126–139, doi: 10.1007/978-3-030-25027-0_9.

[28] G. Vidal, "Efficient Classical Simulation of Slightly Entangled Quantum Computations," *Physical Review Letters*, vol. 91, no. 14, p. 147902, 2003, doi: 10.1103/PhysRevLett.91.147902.

[29] A. Montanaro, "Quantum algorithms: an overview," *npj Quantum Information*, vol. 2, no. 1, p. 15023, 2016, doi: 10.1038/npjqi.2015.23.

[30] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models," in *Proceedings of the Neural Information Processing Systems Datasets and Benchmarks Track*, 2021. [Online]. Available: https://arxiv.org/abs/2104.08663

[31] D. Wadden, S. Lin, K. Lo, L. L. Wang, M. van Zuylen, A. Cohan, and H. Hajishirzi, "Fact or Fiction: Verifying Scientific Claims," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 7534–7550, doi: 10.18653/v1/2020.emnlp-main.609.

[32] V. Boteva, D. Gholipour, A. Sokolov, and S. Riezler, "A Full-Text Learning to Rank Dataset for Medical Information Retrieval," in *Advances in Information Retrieval*, 2016, pp. 716–722, doi: 10.1007/978-3-319-30671-1_58.

[33] M. Maia, S. Handschuh, A. Freitas, B. Davis, R. McDermott, M. Zarrouk, and A. Balahur, "WWW'18 Open Challenge," in *Companion of the The Web Conference 2018 on The Web Conference 2018 - WWW '18*, 2018, pp. 1941–1942, doi: 10.1145/3184558.3192301.

[34] K. Järvelin and J. Kekäläinen, "Cumulated gain-based evaluation of IR techniques," *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002, doi: 10.1145/582415.582418.

[35] National Institute of Standards and Technology, "trec_eval," accessed: Oct. 24, 2026. [Online]. Available: https://github.com/usnistgov/trec_eval

[36] M. D. Smucker, J. Allan, and B. Carterette, "A comparison of statistical significance tests for information retrieval evaluation," in *Proceedings of the sixteenth ACM conference on Conference on information and knowledge management*, 2007, pp. 623–632, doi: 10.1145/1321440.1321528.

[37] S. Holm, "A Simple Sequentially Rejective Multiple Test Procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: https://www.jstor.org/stable/4615733

[38] R. Dror, G. Baumer, S. Shlomov, and R. Reichart, "The Hitchhiker's Guide to Testing Statistical Significance in Natural Language Processing," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2018, pp. 1383–1392, doi: 10.18653/v1/P18-1128.

[39] N. S. Puppala and R. Manoharan, "Trust model to identify reputed miners in blockchain pool mining," in *Proc. Int. Conf. Signal Process., Comput., Electron., Power Telecommun. (IConSCEPT)*, 2023, pp. 1–6, doi: 10.1109/iconscept57958.2023.10170616.

[40] ——, "Cluster-based trust evaluation of blockchain miners in pool mining," in *Proc. 2nd Int. Conf. Adv. Comput. Intell. Commun. (ICACIC)*, 2023, pp. 1–6, doi: 10.1109/icacic59454.2023.10435217.

[41] ——, "Optimizing pool mining performance: a VIKOR-based model for identifying reputed miners in blockchain networks," *Concurrency and Computation: Practice and Experience*, vol. 36, no. 21, p. e8211, 2024, doi: 10.1002/cpe.8211.

[42] N. S. Puppala, "Mitigating malicious activities in blockchain mining: the role of reputation-based protocol," *Security and Privacy*, vol. 8, no. 4, p. e70052, 2025, doi: 10.1002/spy2.70052.

[43] N. S. Puppala and R. Manoharan, "Enhancing mining pool performance and security through optimized cluster-trust consensus mechanism in blockchain networks," *Concurrency and Computation: Practice and Experience*, vol. 37, no. 9–11, p. e70077, 2025, doi: 10.1002/cpe.70077.

[44] N. S. Puppala, M. K. Morampudi, R. K. Suggala, and A. H. L. Swaroop, "Strategic miner selection for optimizing block generation time in PoW-based blockchain pool mining using SMNST framework," in *Proc. OITS Int. Conf. Inf. Technol. (OCIT)*, 2024, pp. 690–695, doi: 10.1109/ocit65031.2024.00125.

[45] S. K. Battina, R. K. Suggala, S. R. Dangeti, M. K. Kumar, P. K. Vadrevu, and N. S. Puppala, "Secure pool mining through SVM-based miner classification and computation validation in blockchain networks," in *Proc. OITS Int. Conf. Inf. Technol. (OCIT)*, 2024, pp. 554–559, doi: 10.1109/ocit65031.2024.00102.

[46] P. L. Sai, K. G. Venkat, N. S. Puppala, T. M. R. Shankar, L. P. V. A. Lanka, and T. Bharadwaja, "Detecting marine plastic waste using aerial and satellite images: a multimodal fusion approach," in *Proc. 1st Int. Conf. Data Sci. Intell. Netw. Comput. (ICDSINC)*, 2025, pp. 42–47, doi: 10.1109/icdsinc66221.2025.11448230.

[47] P. N. Sravanthi, P. V. L. Vardhan, G. V. Byraju, and S. Annamneedu, "Incident-aware reinforcement learning for automated blockchain smart contract vulnerability detection and repair," in *Proc. 9th Int. Conf. Comput. Methodol. Commun. (ICCMC)*, 2026, pp. 871–877, doi: 10.1109/iccmc69250.2026.11625095.

[48] S. Bhimineni, C. S. Susmitha, H. Korukonda, E. S. Kandimalla, and N. S. Puppala, "Fraud detection in financial transactions: a rule-based and machine learning approach," in *Proc. Int. Conf. Emerg. Trends Signal Process. Comput. Intell. (ICCSPCI)*, 2025, pp. 1–7, doi: 10.1109/iccspci68199.2025.11451569.

[49] S. H. V. Chikkala, T. R. Tumma, P. N. Sravanthi, and L. V. S. Paila, "Hybrid deep learning pipeline with CNN-based feature embeddings and dimensionality reduction for plant disease detection," in *Proc. 1st Int. Conf. Data Sci. Intell. Netw. Comput. (ICDSINC)*, 2025, pp. 735–740, doi: 10.1109/icdsinc66221.2025.11448138.

[50] G. Varshithachowdary, M. Yasaswini, M. Meghana, and N. S. Puppala, "A hybrid machine learning and deep learning approach for traffic volume classification and forecasting," in *Proc. 1st Int. Conf. Data Sci. Intell. Netw. Comput. (ICDSINC)*, 2025, pp. 58–63, doi: 10.1109/icdsinc66221.2025.11448208.

[51] N. S. Puppala, U. V. S. S. S. Rohith, R. P. Moganti, and S. K. Chebiyyam, "Visionary harvest: integrating custom blur layers into VGG19 for obscured edible classification," in *Proc. 9th Int. Conf. Comput. Methodol. Commun. (ICCMC)*, 2026, pp. 1548–1553, doi: 10.1109/iccmc69250.2026.11625194.

[52] A. Javadi-Abhari *et al.*, "Quantum Computing with Qiskit," arXiv:2405.08810, 2024. [Online]. Available: https://arxiv.org/abs/2405.08810

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